

Current Status of the Simulation of Radiative Effects with McMule

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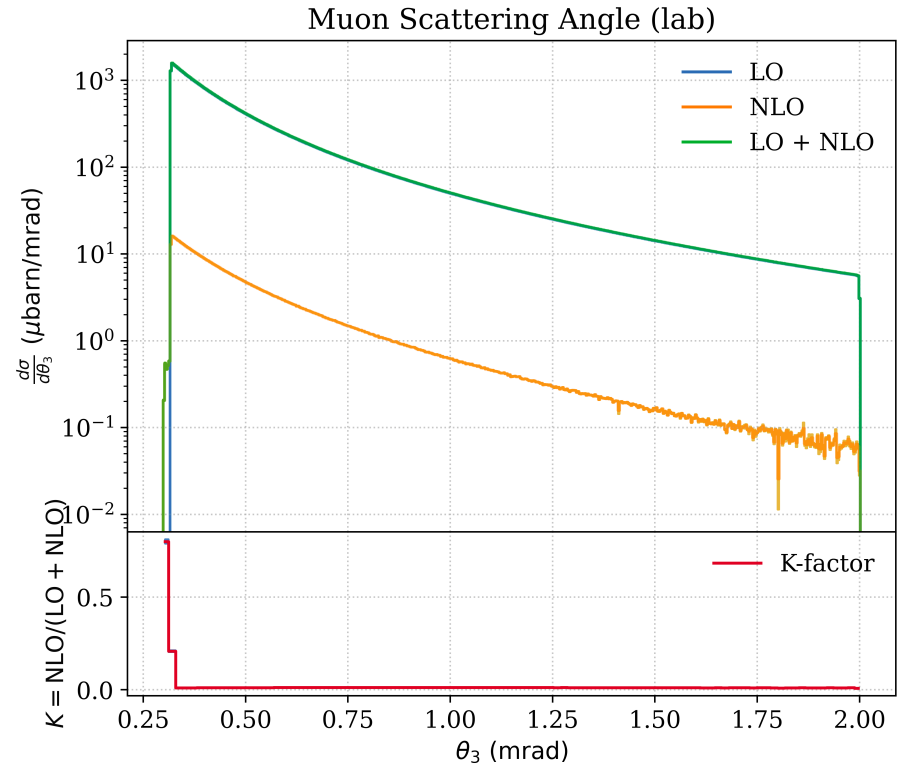
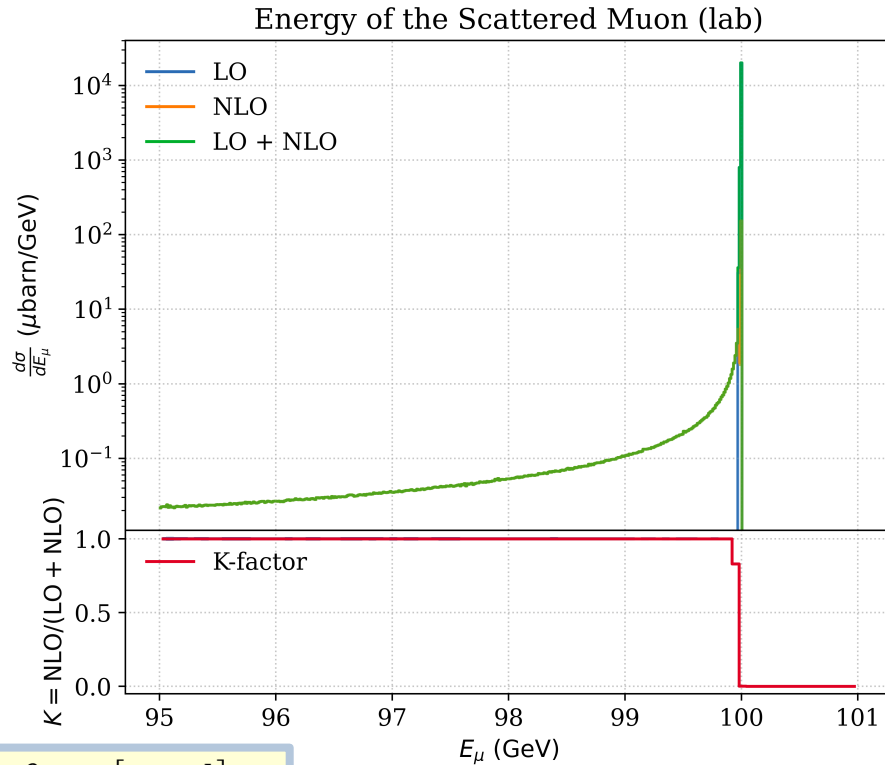


McMULE

Project Overview and Progress

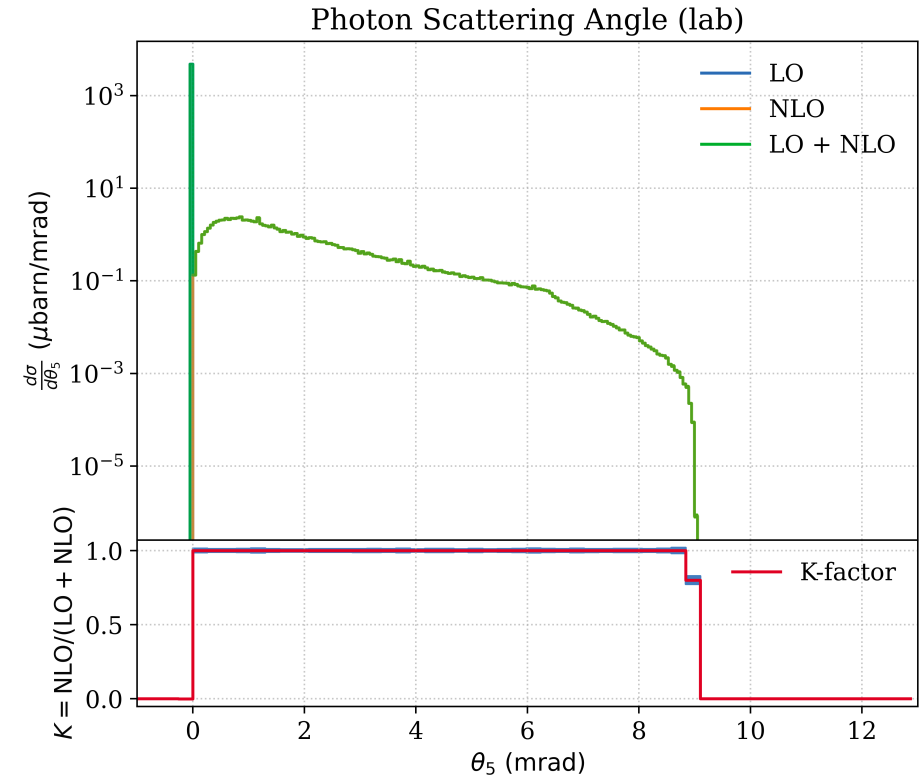
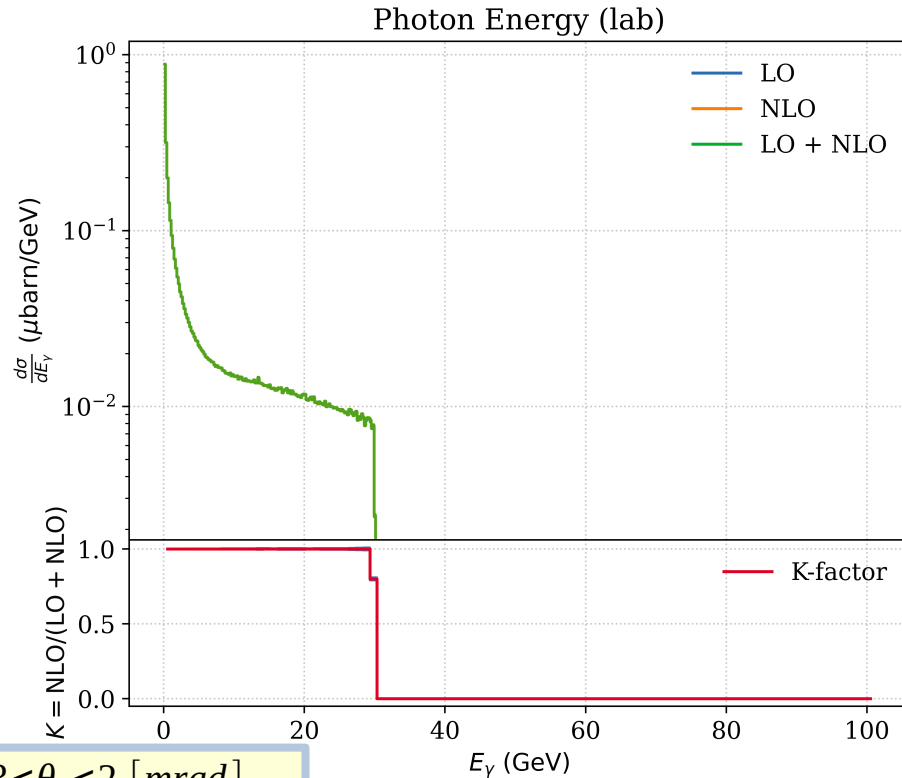
- Topic: Simulation of Radiative Effects for the AMBER Proton Radius Measurement with McMule
- Learning to work with McMule with support from Marco Rocco
- Simulation of LO ($\mu + p \rightarrow \mu + p$) and NLO processes
- Especially investigating the real-photon emission process
- Investigation of the impact of different cuts on the kinematic
- Cuts used to obtain the results in this presentation:
 - $0.3 < \theta_\mu < 2. [mrad]$
 - $E_\mu > 70 \text{ GeV}$
 - $-12. < \theta_\gamma < 12. [mrad]$

Muon: Energy & Scattering Angle



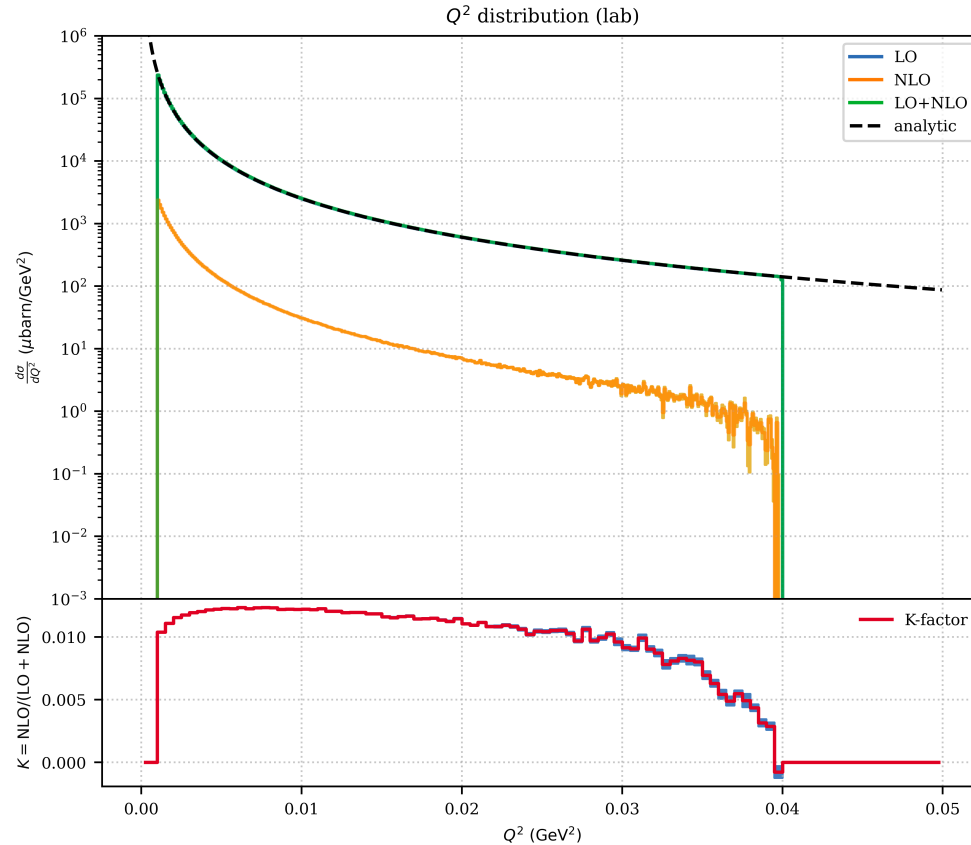
$0.3 < \theta_\mu < 2. [\text{mrad}]$
 $E_\mu > 70 \text{ GeV}$
 $-12. < \theta_\nu < 12. [\text{mrad}]$

Photon: Energy & Scattering Angle



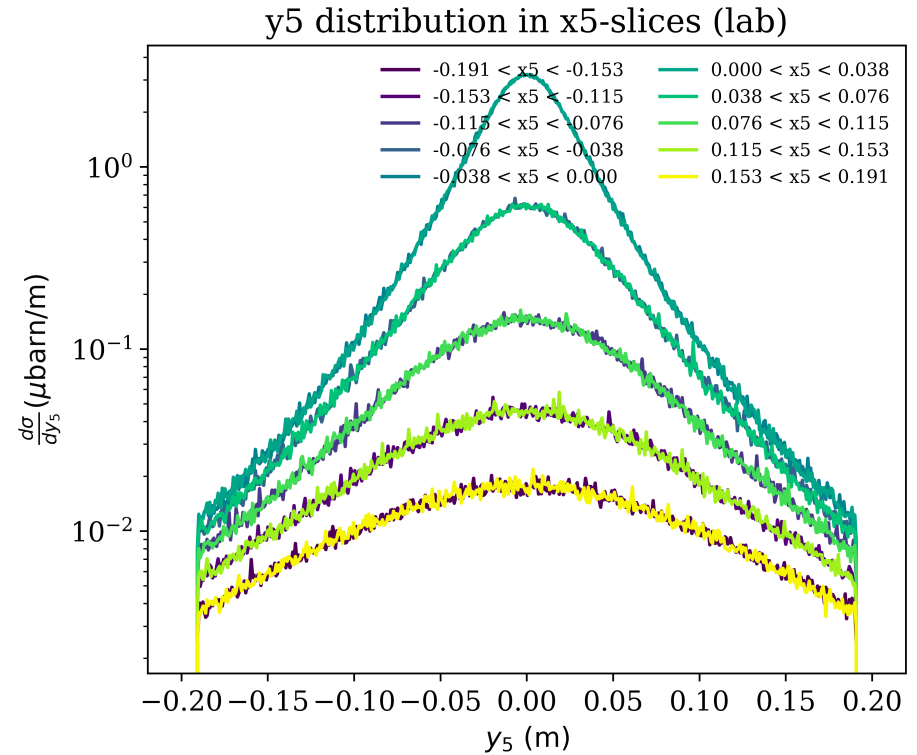
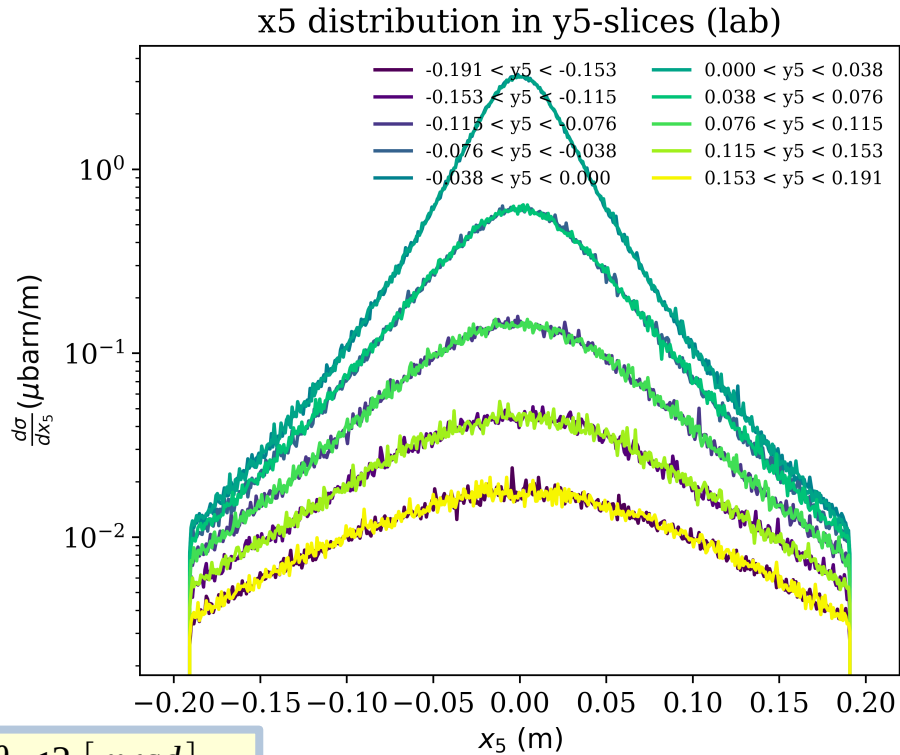
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Momentum Transfer



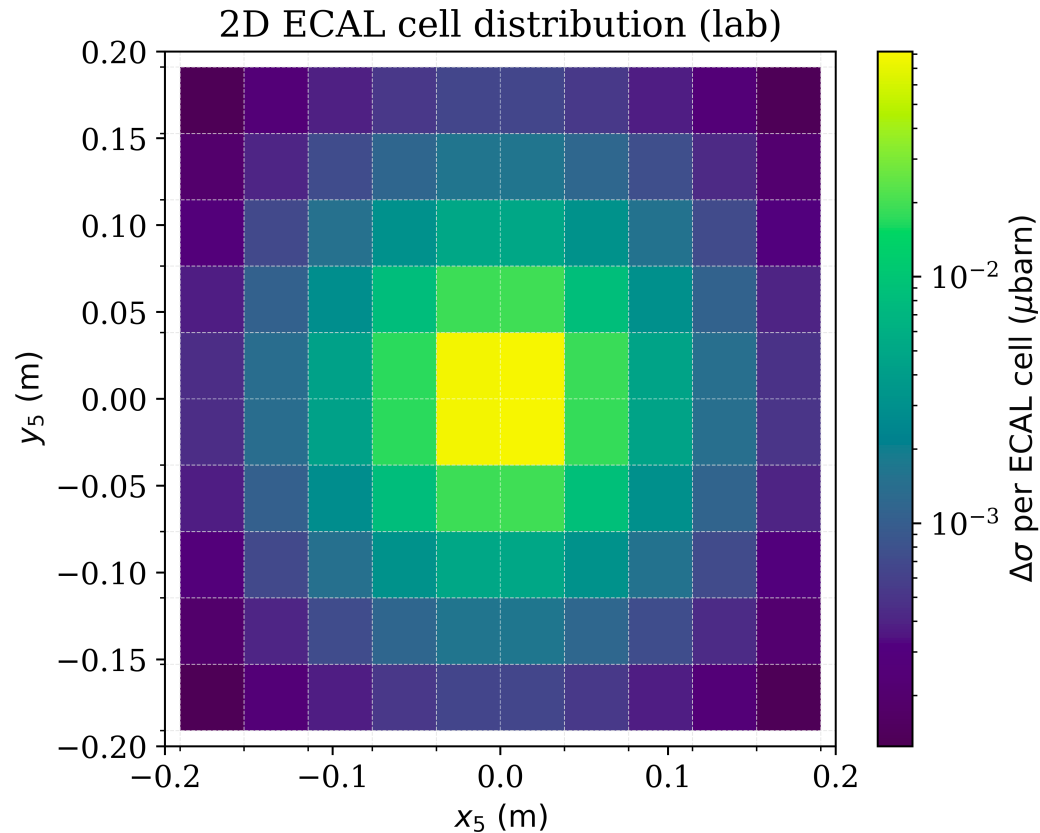
$0.3 < \theta_\mu < 2. [\text{mrad}]$
 $E_\mu > 70 \text{ GeV}$
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Where Photons Hit the ECAL



$0.3 < \theta_\mu < 2. [\text{mrad}]$
 $E_\mu > 70 \text{ GeV}$
 $-12. < \theta_\gamma < 12. [\text{mrad}]$

Expected Photon Cross Section on ECAL



$0.3 < \theta_\mu < 2. [\text{mrad}]$
 $E_\mu > 70 \text{ GeV}$
 $-12. < \theta_\gamma < 12. [\text{mrad}]$

Expected Cross Sections and Count Rates

$$0.001 < Q^2 < 0.04 [GeV^2/c^2]$$

Photon Energy	100 MeV	200 MeV	500 MeV
σ_{LO} [mb]	0.250567	0.250567	0.250566
R_{LO} [1/s]	42.53652	42.53653	42.53642
$\sigma_{R\gamma}$ [mb]	0.000091	0.000080	0.000082
$R_{R\gamma}$ [1/s]	0.0154	0.0136	0.0139
σ_{NLO} [mb]	0.00284	0.00283	0.00283
σ_{FULL} [mb]	0.253407	0.253396	0.253397

Count rate input values:

Target length: 80 cm

Hydrogen pressure: 20 bar

2.687×10^{19} protons/cm³

Beam intensity: 2×10^6 1/s

$$0.3 < \theta_{\mu} < 2. [mrad]$$

$$E_{\mu} > 70 GeV$$

$$-12. < \theta_{\gamma} < 12. [mrad]$$

- The number of integration points (Runtime) has a significant impact on the result and its accuracy! → Under Investigation
- The real-photon process contributes at the ~0.03% level

BACKUP

Used Equations

- def calculate_rate(sigma_mb):
 ltarget = 80 #cm
 Hpres = 20 #bar
 Npvol = 2.687*1e19 #Protons/cm^3
 lbeam = 2*1e6 #1/s
 Np = Npvol * 2 * (Hpres/1.013) * ltarget #Protons/cm^2 (target thickness)

 rate = sigma_mb * 1e-27 * Np * lbeam
 return rate #1/s